

Review

A competition model of multilineage priming and cell-fate decisions

Tobias Steinschaden,¹ Louis Faure,² Ruslan Soldatov,² and Igor Adameyko^{1,3,*}
¹Department of Neuroimmunology, Center for Brain Research, Medical University Vienna, 1090 Vienna, Austria

²Department of Epidemiology & Biostatistics, Memorial Sloan Kettering Cancer Center, New York, NY 10065, USA

³Department of Physiology and Pharmacology, Karolinska Institutet, 17177 Stockholm, Sweden

*Correspondence: igor.adameyko@meduniwien.ac.at
<https://doi.org/10.1016/j.celrep.2025.116690>

SUMMARY

The transition from a single cell to a body of trillions requires each cell to make fate decisions, committing to one identity among many possible options. This review aims to examine how progenitors and stem cells generate diverse fates through multilineage priming and the competitive interplay of biasing gene-expression programs. Microheterogeneity among progenitors, combined with differential fate biases, gives rise to cell populations with collective multipotency, an emergent state in which the population self-regulates to balance the production of all downstream fates during embryogenesis. We discuss how progenitor cells navigate the epigenetic landscape to control descendant cell-type proportions and how multilineage priming helps reconcile inductive, communication-driven modes of development with autonomous, self-intrinsic fate selection. In parallel, we review computational approaches that leverage single-cell transcriptomics, clonal lineage tracing, and multiomics integration to reconstruct developmental trajectories, identify biases, and dissect the mechanisms of fate selection and commitment.

CHOICE VIA COMPETITION OF OPPOSING GENE-EXPRESSION PROGRAMS: MULTILINEAGE PRIMING

Embryonic development proceeds through a series of fate decisions, whereby cells gradually restrict their differentiation potential and commit to specific lineages.^{1–4} We use the term cell fate to talk about the trajectories that lead to and including the cell types that a cell is differentiating toward. The progenitor cells, especially those in early development, are often multipotent, meaning they have the opportunity to generate multiple cell types, defined as “potency.” They explore the dynamically unfolding environment together with their daughter cells to gradually narrow down the scope of their potency and give rise to specific subsets of adult cell types. This process requires integrating multiple signals and intrinsic cell states over time. Classic observations, such as the divergence of trophectoderm and inner cell mass in the mammalian blastocyst, illustrate this process.⁵ These decisions are not preordained—development remains flexible across a range of circumstances, including the unexpected loss of embryonic parts.^{6,7} This is achieved by taking advantage of cell-cell communication including cell-fate biasing signals.^{8–10} In this review, we focus on how biasing experienced by progenitor cells during development is studied, especially using single-cell methods. Contemporary investigations of cell fate using single-cell and spatial transcriptomics rest atop a methodological evolution, ranging from early 20th century dye labeling and later tissue-grafting experiments over genetic lineage-tracing studies, and were most recently enabled by advances in readouts such as real-time

quantitative PCR (qPCR), bulk transcriptomics, and epigenetic profiling.¹¹

Historical frameworks, such as Waddington’s epigenetic landscape, conceptualize development as a branching trajectory shaped by gene interactions¹² (Figure 1A). With the rise of single-cell technologies, it has become clear that developmental transitions are continuous, not discrete, which helps us model their paths on the epigenetic landscape.^{13,14} Adult terminal cell types appear discrete only due to the loss of intermediate progenitor states, much like species in evolution^{15,16} (Figure 1B). For the human reader, the progenitor cells can still be separated from each other in agreement with the continual paradigm (similar to how we define the colors of the spectrum), as well as our communal consensus about how much difference is enough to claim another cell type or cell state.^{17,18} Of note, their separation will be, just like the spectral and species evolution analogies, an arbitrary cutoff between progenitor types. Therefore, this discussion reflects the issues of scientific communication because cells can be mathematically described, defined, and plotted using unbiased tools based on single-cell multiomics readouts. The precision of the computational approach might alleviate the problem of our language to define a position of a cell somewhere along the continuity of cell-lineage trajectories during development. Indeed, any cell can be represented as a coordinate in a multidimensional transcriptional space or on a differentiation trajectory. In the case of dynamic data—for instance, based on RNA velocity or metabolic labeling approaches^{19,20}—the coordinate can become vectorized. In addition to this, transient cell types might be objectively defined via



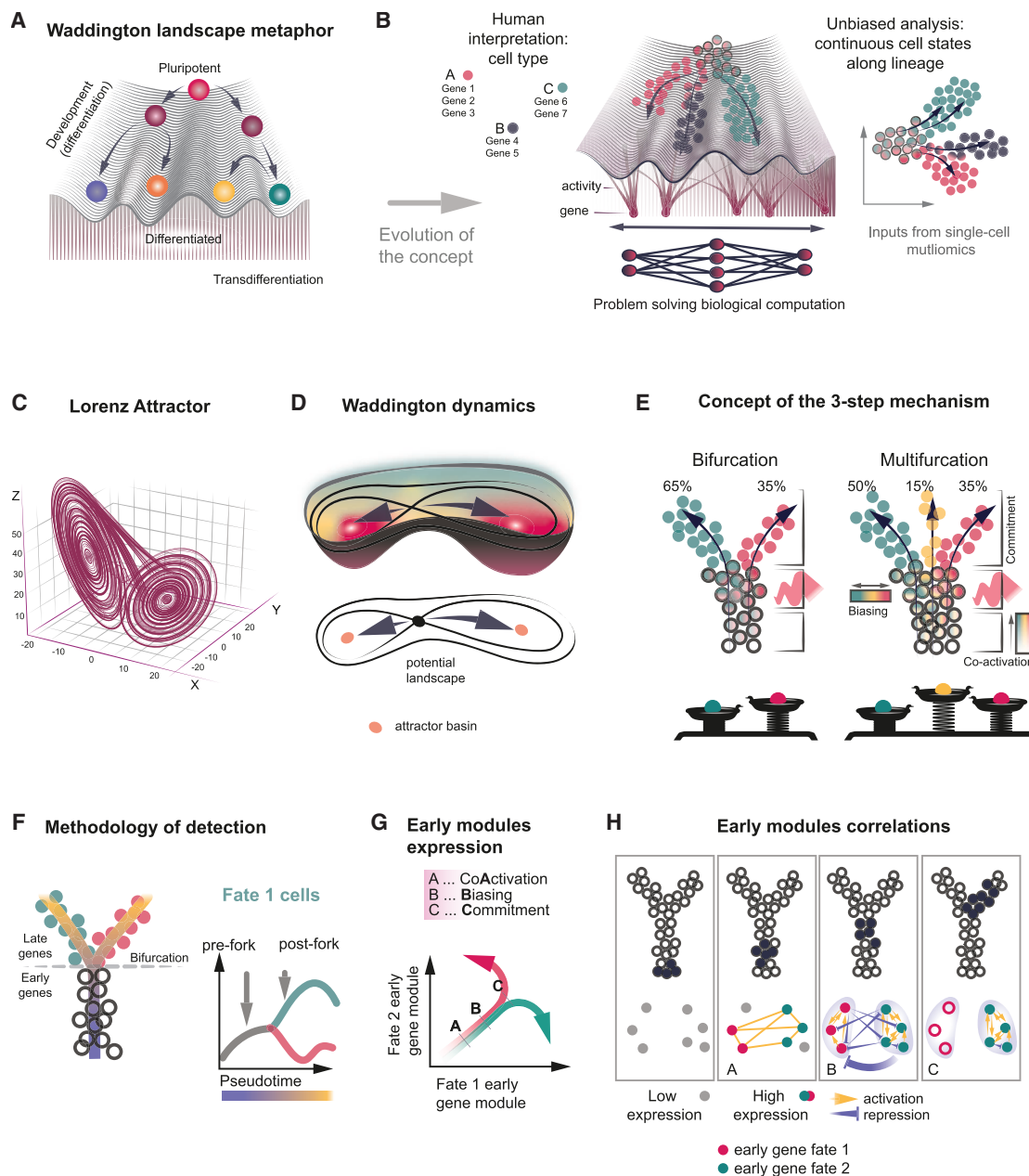


Figure 1. Conceptual frameworks for understanding cell-fate decisions

(A) Waddington's epigenetic landscape: a cell (ball) rolling down a hill toward alternative fates.

(B) Beneath the landscape lies a vast regulatory network shaped by evolutionary feedback, reminiscent of an artificial neural network. Researchers attempt to categorize points, but the landscape's continuity complicates this. Dimensionality reduction enables partial categorization while preserving continuity.

(C) Lorenz attractor: a deterministic system governed by known equations, yet chaotic and unpredictable without precise initial conditions. Despite this, solutions converge on deterministic attractor states.

(D) Basins of attraction (3D and top-down views): sets of states that converge to a stable attractor (orange dots). Basin topology can be reshaped by external signals and epigenomic mechanisms.

(E) Single-cell RNA-seq reveals opposing early transcriptional modules. Along pseudotime, early co-activation gives way to rapid divergence of gene expression. (F) Three-step model of bifurcation/multifurcation: co-activation, biasing, and fate commitment. Fate proportions reflect intrinsic biases (scales) and extrinsic cues (pink wave).

(G) Early gene module dynamics: A, co-expression of both programs; B, relative upregulation of one program; C, repression and loss of the opposing program. (H) Fate-specific gene expression along bifurcation: co-activated modules (green and red dots) reinforce themselves while repressing each other. Eventually, some cells commit to one program (green dots) and extinguish the opposing module (red dots), establishing new identity.

their lineage tree structure: the segments of the tree between bifurcations (fate splits) might be rendered as cell types in an unbiased way. Also, in colloquial language, the difference between a cell type and a cell state may reflect the stability of a cell phenotype over time. Cell types in this scheme might represent rather stable or prolonged situations, whereas cell states fluctuate, oscillate, and appear as temporal responses to stimuli. Thus, one cell type can encompass multiple partially overlapping cell states.

Divergent cell states arise from differential gene expression governed jointly by transcription factors (TFs), chromatin states, and 3D genome architecture.²¹ While the “master regulator” concept originally described TFs at the top of regulatory hierarchies such as Pax6 in eye development, its modern usage has become either overly broad or narrowly restrictive.^{22–24} Pax6, for instance, is expressed in many tissues and species but only induces eyes under specific conditions, depending on local co-factors.^{25–29} This issue is resolved by a more functional model, which is offered by gene-regulatory networks (GRNs), which organize TFs and chromatin regulators into logic-based and contextual subcircuits.^{30–33} First championed by Eric Davidson, GRNs reflect how complexity emerges from combinatorial and modular logic.³⁴

Because GRNs adapt to external states, they unfold as dynamic processes rather than static hierarchies, and new technologies have become essential for tracing their evolution in real time.³⁵ The accumulating advances in single-cell transcriptomics and multiomics have enabled high-resolution mapping of developmental trajectories and TF activation at specific points, helping us to understand the gradual transformations of GRNs in time.^{36,37} By reconstructing pseudotime paths—a computationally inferred direction by which cells are ordered in a continuous progression of gene expression, which approximates differentiation—we infer trees based on transcriptional similarity, which show bifurcations of transcriptional states where cell-fate decisions occur.³⁸ The transcriptional trees show not only bifurcations and fate splits, which were known from before, but also project the transcriptional states of the cells on the trajectories that bifurcate or assume more complex topology (number of branches and hierarchical nodes) depending on the scale of the overall reconstructed tree.³⁵ These topologies might also include multifurcations and fate convergences. Though not fully equivalent to clonally defined lineages, these trees built on plotting gradual transcriptional changes are invaluable for understanding mechanisms underlying fate selection just prior to commitment, as they provide a testable working hypothesis.^{39,40} Integration of additional clonal information greatly improves the interpretability of the overall lineage trees with overlaid transcriptional components, helping to pinpoint the cases of converging trajectories of cell differentiation and decipher the factors influencing the results of fate decision-making.^{41–45} Also, clonal information helps to better resolve and define cell types, especially in cases of transcriptionally similar cells such as fine neuronal subtypes.⁴⁵

To move beyond descriptive analysis of trees, several groups have applied mathematical and physical concepts to model cell-fate selection as a dynamic system. Mojtahedi et al.⁴⁶ and other researchers^{47,48} explored the idea of gene-expression “attrac-

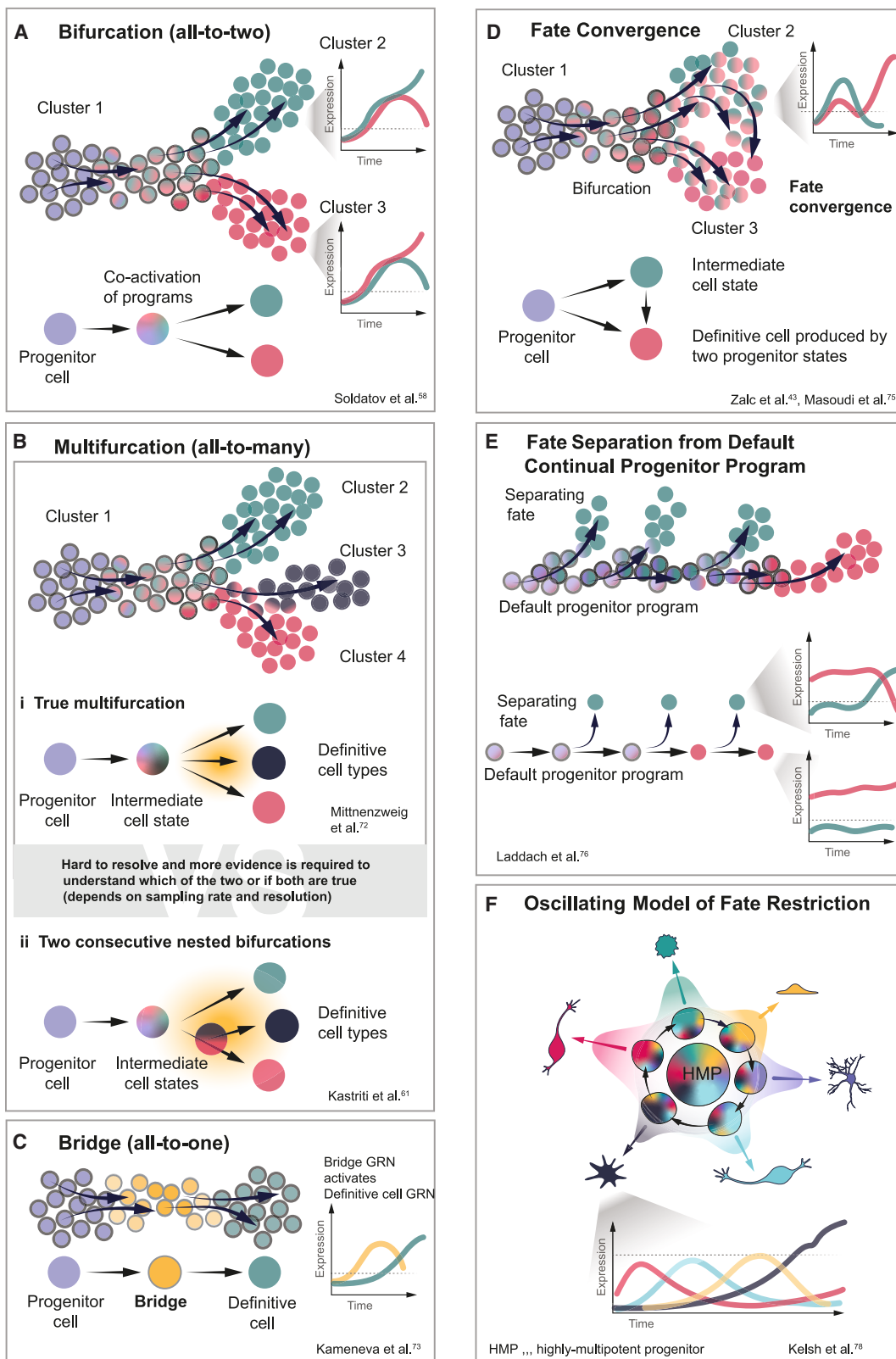
tors,” stable progenitor states maintained by GRNs (Figure 1C). Such “attractors” can be reminiscent of Lorenz attractors, which approximate chaotic dynamics, yet they arise from deterministic differential equations.^{49,50} Fate transitions may occur when conceptually analogous intracellular attractors destabilize, allowing cells to shift into new states.^{51–53}

Using single-cell real-time qPCR, Bargaje et al.⁵⁴ applied a similar framework to human induced pluripotent stem cells differentiating into cardiomyocytes, identifying a bifurcation point between mesodermal and endodermal fates marked by elevated transcriptional heterogeneity. Also, Sáez et al.^{55,56} took advantage of catastrophe theory and dynamical systems to reinterpret Waddington’s landscape, showing how GRNs and external signals govern cell-fate transitions, including stochastic deviations from canonical paths (Figure 1D).

Kang et al.⁵⁷ applied models from statistical mechanics to stem cell differentiation, showing that fate transitions resemble phase changes in physics, being driven by gene-expression noise and intercellular communication. Their Boolean network model demonstrated how progenitor populations self-organize into stable differentiated states through symmetry-breaking events.

Parallel to these theoretical frameworks, the Adameyko and Kharchenko laboratories inferred a less abstract three-step mechanism of fate selection based on the analysis of single-cell data obtained from multipotent neural crest cells.⁵⁸ This was enabled by the methodological improvements of Smart-seq2, which provides full-length transcripts and high coverage.⁵⁹ First, co-activation of opposing biasing gene modules occurs in the early progenitors (Figure 1E), representing a case of multilineage priming. Next, these biasing modules interact antagonistically, leading to divergence of cell transcriptomes (Figures 1F and 1G). Finally, one module dominates, repressing the alternative and stabilizing commitment to a specific fate (Figure 1H). Importantly, early biasing does not necessarily involve mature fate-determining TFs; instead, subtle interactions between signaling pathways, receptor expression, and chromatin states might guide initial biasing before terminal identity programs are fully established.^{58,60} Thus, the early modules, rather than acting through master regulators, may represent weak and overlapping fate determinants.^{58,61} Commitment results when one competing GRN matures more rapidly, suppressing the rival and locking in a fate. Thus, lineage progression involves sequential binary choices governed by co-activation of biasing programs, their competition, and commitment.^{58,62,63} This framework emphasizes that in a context of inductive development, fate decisions are probabilistic, shaped by extrinsic cues and/or intrinsic history, not hardwired switches.^{58,60,63,64} For every developing cell lineage, the combination of external biasing signals and intrinsic states might be different or even unique. After bifurcation, branch-specific genes emerge, including GRN components required for differentiation and maintenance of chosen fates.^{58,65,66} Often, master regulators are among these late genes, acting more to stabilize pre-established fates than to direct initial biasing.⁶¹ These post-commitment genes ensure trajectory fidelity and solidify cell identity.

The general concept of multilineage priming and competing programs driving fate selection is supported by a plethora of independent evidence. For example, myeloid fate choice involves



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early co-expression of antagonistic factors PU.1 and C/EBP.⁶⁷ These initiate mutually repressive downstream circuits (Egr/Nab vs. Gfi1), leading to resolution of fate conflicts. Once one program gains dominance, it represses the alternative, stabilizing identity and preventing reprogramming. Such a dynamic has been modeled *in silico* by a six-component GRN motif called “alternate subcircuits driven by reciprocal repressors.”³⁰

In hematopoietic stem cells (HSCs), multilineage priming seems to be common.⁶⁸ Early transcriptional programs include genes from multiple lineages, supporting the capacity for flexible fate choices.^{63,69} Cell-sorting experiments showed that early expression of general fate markers precedes expression of lineage-specific regulators.⁶⁸ A similar phenomenon occurs in Lgr5⁺ intestinal stem cells, where early progeny co-express absorptive and secretory lineage markers.⁷⁰ Over time, these programs become refined, resolving into distinct cell types. In cardiac progenitors, early mixed-lineage expression enables adaptation to complex morphogenetic signals, eventually producing the heart’s diverse cell types.⁶⁹ In nephrogenesis, progenitors express markers for multiple nephron segments before commitment.⁷¹

To analyze the dynamics of such multilineage priming computationally, the tool scFates was developed.⁶⁰ It identifies early and late gene modules from single-cell transcriptomics trajectories and traces their divergence relative to bifurcation points. By correlating early modules backward along the progenitor trajectory, scFates can infer candidate biasing programs and their interactions.

DIVERSE MODELS OF FATE RESTRICTION IN THE CONTEXT OF A THREE-STEP MECHANISM

In this section, we discuss how the competition-based and multilineage priming-based three-step mechanism can account for a range of the cell-fate decision models that have been put forth in the past. The three-step model applies broadly but admits variation. While many decisions resemble bifurcations (Figure 2A), some resemble multifurcations⁷² (Figure 2B) or bridges,⁷³ where cells choose between continuing as progenitors or differentiating (Figure 2C). Others involve fate convergence^{43,74,75} (Figure 2D), whereby different lineages generate similar cell types. All these dynamics reflect varied developmental logic and are observed in different datasets, including the examples of gastrulation⁷² and progression of the neural crest lineage.^{58,61}

In some systems, one fate pathway acts as the “default,” while an opposing program is activated *de novo* (Figure 2E). For instance, during enteric nervous system development, glial fate appears retained from the progenitor branch, while the neuro-

genic program is selectively activated.⁷⁶ A similar principle likely operates in stem cell niches, where asymmetric division yields one differentiating daughter and one self-renewing cell that maintains the default program.⁷⁷

While bifurcation models explain a lot, some studies propose more complex oscillatory or dynamic circuits driving fate⁷⁸ (Figure 2F). Recently, the phenomenon of a multilineage priming and the probabilistic nature of fate selection inspired the “cyclical fate restriction” (CFR) fate-selection model, which originated from the analysis of neural crest and pigment cell lineage in zebrafish.^{78,79} CFR proposes that neural-crest-derived highly multipotent progenitors do not follow a linear trajectory toward a specific fate. Instead, these cells cycle through different transcriptional substates, each biased toward a particular fate, such as a melanocyte or iridophore, without becoming fully committed. This cyclical model suggests that the commitment to a single fate may also occur only after several rounds of oscillating biasing, depending on the substate.^{78,79} Biologically, this might be driven by fast spatiotemporal changes in the cell microenvironment, for instance during cell migration or tissue remodeling. This cyclical model expands the more traditional “direct fate restriction” (DFR, Figure 3A) and “progressive fate restriction” (PFR, Figure 3B) models.⁷⁹ To provide more context, the PFR model posits that progenitors sequentially restrict their fate potential through a series of intermediate, less multipotent stages before final fate commitment.^{80,81} In theory, this process can be seen as a hierarchical progression whereby multipotent progenitors gradually lose their potential to differentiate into various cell types, eventually becoming committed to one specific lineage. The DFR model, quite differently, suggests that neural crest cells maintain their multipotency until they receive specific environmental signals that direct them to adopt a particular fate directly, without passing through intermediate stages of potency reduction. In principle, the CFR model could reconcile observations that underpin both the PFR and DFR hypotheses by providing a rather dynamic understanding of fate specification (Figure 3C).

Importantly, CFR, PFR, and DFR models are all compatible with the three-step mechanism,⁵⁸ which in the case of CFR does not reach the commitment stage for some time and oscillates at the co-activation and biasing stages (Figure 2F). Also, in this case it might involve more than two competing programs at the same time and in the same cell. However, all of this requires further experimental investigation in a broad range of systems.

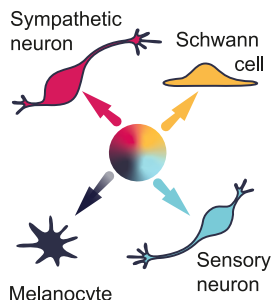
The probabilistic and highly dynamic nature of biasing suggests that there might be a potential for rebiasing and reversions

Figure 2. Various types of cell-fate transitions and their relation to the three-step mechanism

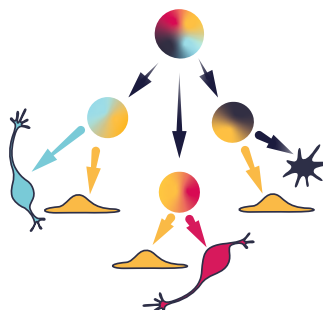
- (A) Bifurcation: one progenitor population splits into two downstream fates.
- (B) Multifurcation: one progenitor gives rise to more than two fates. Whether true multifurcations exist in development or represent rapid consecutive bifurcations remains unresolved.
- (C) Bridge transition: one progenitor passes through an intermediate state before producing a single downstream fate.
- (D) Fate convergence: downstream fates may retain plasticity, enabling switches to an alternative fate after bifurcation.
- (E) Default path separation: cells follow a default differentiation path unless extrinsic cues trigger an alternative program that antagonizes the default. This mechanism recurs throughout development and even in adult cells.
- (F) Cyclical fate restriction (alternative model): cells cycle through poised states that allow direct acquisition of specific fates.

Diverse Models of Cell Fate Restriction

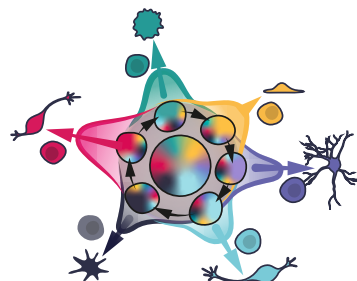
A Direct Fate Restriction



B Progressive Fate Restriction



C Cyclical Fate Restriction



Intrinsic Biasing Factors Can Override Extrinsic Cues

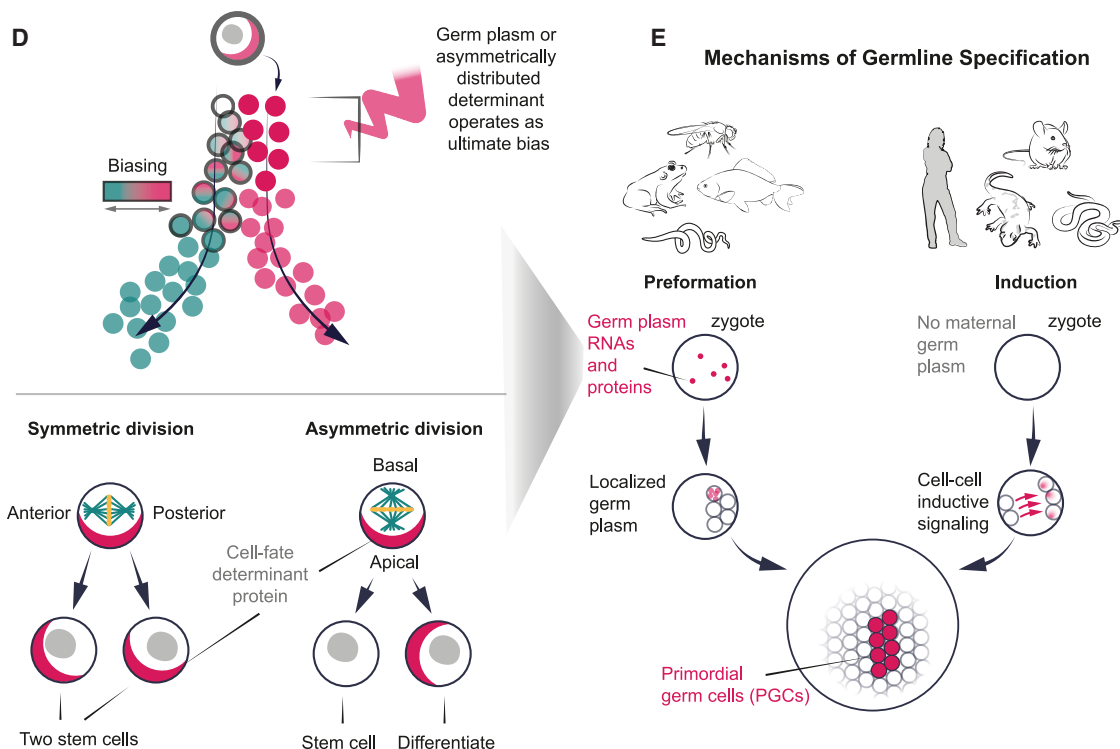


Figure 3. Models of cell-fate restriction and intrinsic biasing factors

(A–C) Diverse models of cell-fate restriction. (A) Direct fate restriction: cells commit from a multipotent progenitor to a fate. (B) Progressive fate restriction: cells commit from a multipotent progenitor toward a lineage that comprises a subset of the potential fates. They progressively lose potency while developing toward different fates. (C) Cyclical fate restriction: multipotent progenitors cycle through a defined series of transcriptionally poised states that allow them to commit—contextual cues permitting—to a specific fate directly. Otherwise, the progenitor continues to the next poised state.

(D) Two ways of conceptualizing an asymmetrically distributed cell-fate determinant (e.g., germ plasm) overriding any inductive signals: one in the context of the three-step mechanism (upper) and the other the context of orientation of cell divisions (lower).

(E) Intrinsic vs. inductive development in the example of germline specification.

from a more progressive state to a less progressive state, which follows what the CFR model postulates.⁷⁸ This logic might apply to some multicellular systems but not others, and understanding the distribution of CFR, PFR, and DFR models in nature will

become a subject of serious future efforts. This “temporal” flexibility most likely ends with the commitment to a specific fate and cell-type manufacturing process, except for the systems specifically wired for reversals—for example, during tissue or organ

regeneration. Indeed, in the regenerating salamander limb, fully differentiated skeletal, muscle, and soft connective tissue cells actively dedifferentiate to form a regenerative blastema, a conglomerate of reverted progenitor-like cells that will recreate the limb during processes of lineage development partly resembling the progression of an embryonic limb bud.^{82,83}

On a separate note, although the classic three-step mechanism describes the cases of inductive lineage development (as postulated for the neural crest lineage), it can theoretically operate under intrinsic programming conditions (for an explanation of inductive vs. intrinsic development, see [Figures 3D and 3E](#)). Our original hypothesis suggests that, for instance, the molecular determinants of a fate, which are transmitted only to one daughter cell as a part of asymmetric cell division, might function as strong overpowering biases, which always win during the competition of opposing biasing programs.^{84–87} The extreme strength of such biases administered by molecular determinants transmitted in a directional way makes a competition aspect weak and probably non-operational, though not entirely non-existent.

Interestingly, the organisms known to develop autonomously, such as nematodes and tunicates,^{84,86} never execute the entire development in a fully autonomous mode, because later in development (or even simultaneously) they transition to fate selection based on cell-cell interactions.^{84,86} In most cases, the autonomous mode of fate selection due to asymmetry of cell division or some other physical aspects of symmetry breaking is blended with inductive development of other lineages in the same developing embryo. Therefore, although autonomous and inductive modes of development might appear vastly different in terms of molecular mechanisms of fate selection, they co-exist and may possibly transition into each other. It seems that the same three-step mechanism can accommodate both modes, being strongly manipulated by asymmetrically distributed biasing determinants in the case of autonomous development and operating in a true competitive manner with rather subtle biases in the case of the inductive mode. Thus, the true difference between autonomous and inductive development might be rooted in the way biases are introduced and in the variation of their strength.

ROLE OF POISED AND BIVALENT ENHANCERS IN MULTILINEAGE PRIMING AND COMPETITION-BASED FATE SELECTION

The gradual upregulation of biasing and opposing transcriptional programs in the same progenitor cells prior to fate selection requires fast responses from the chromatin structure, which calls the exact modes of chromatin dynamics into question, especially during multilineage priming. The readiness of enhancers and promoters for a fast response during co-assembly of two or more parallel and competing GRNs might be a strategy to ensure a timely choice and proper competition.

Not surprisingly, enhancers are often found to be epigenetically primed in a “poised” state, which ensures that cells are ready to respond quickly to developmental cues.⁸⁸ This poised state is marked by specific histone modifications (such as H3K4me1 without H3K27ac), which do not activate gene expression immediately but prepare the enhancer for future activa-

tion.^{89,90} Some enhancers and promoters exhibit “bivalent” marks, both activating (H3K4me3) and repressing (H3K27me3) histone modifications.^{91,92} This duality is thought to keep key developmental genes in a ready state for activation while simultaneously preventing their premature expression.⁹³ Bivalent domains are particularly notable in stem cells and early developmental stages, providing a flexible mechanism for maintaining and resolving lineage specification.

For example, Minoux et al.⁹⁴ found that early postmigratory cranial neural crest cells across different craniofacial regions show similar chromatin accessibility patterns, yet their gene-expression profiles varies significantly. In particular, genes essential for determining positional identity in cranial neural crest cells exhibit bivalent chromatin marks—both activating (H3K4me2) and repressing (H3K27me3)—at their enhancers and promoters. This bivalency suggests that these genes are poised for activation but require specific signals to commence expression. This study highlights the role of the Polycomb repressive complex, specifically the enzyme Ezh2, in establishing and maintaining these bivalent chromatin states.⁹⁴ Thus, the epigenetic landscape of neural crest cells is primed to respond to local cues that later specify the identity of different craniofacial components.

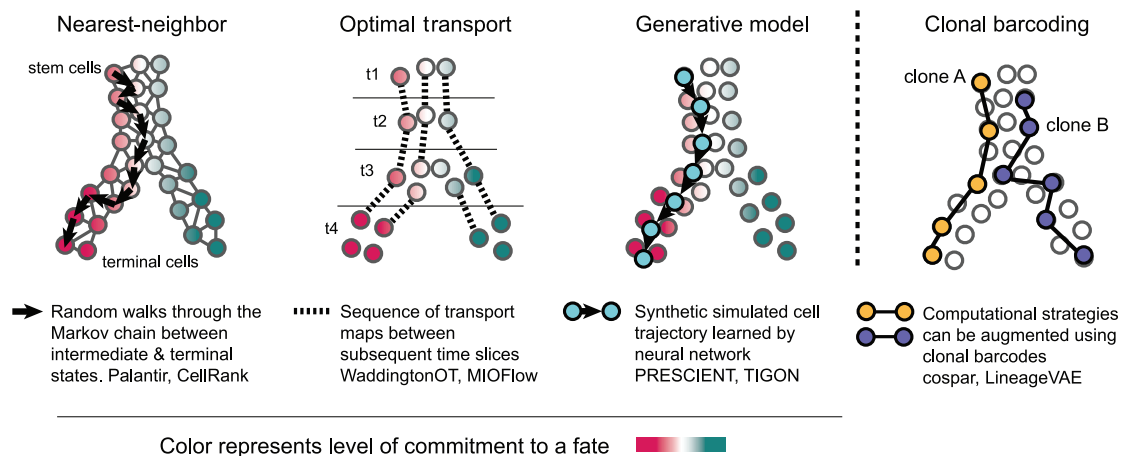
The timing of bivalent and poised enhancer activation is crucial. For example, enhancers that control key developmental genes can become poised early in development, well before the genes are needed.⁹⁵ This allows for rapid gene activation in response to developmental signals or during the assembly of competing gene-expression programs, overall ensuring timely cell differentiation.⁹³ The transition from a bivalent to active or repressed chromatin state often coincides with lineage commitment⁹² and must become an integral part of a three-step mechanism or other fate-restriction model (accounting for a possible diversity of fate-selection strategies), because such bivalency of a genome enables fast gene-expression activation during intracellular competition of opposing GRN components. Later, as pluripotent/multipotent cells make choices and begin to differentiate, bivalent markers at lineage-specific genes resolve into active marks in cells where the gene must be expressed stably and into repressive marks where it should not.⁹²

PROBABILITY OF CHOICE AND MICROHETEROGENEITY AFFECTING FATE SELECTION

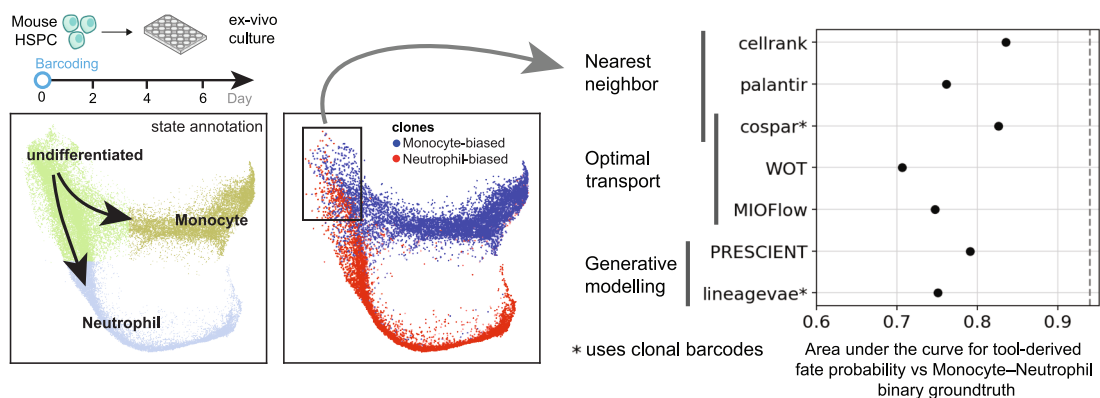
If the goal of development is to generate complexity from simplicity, the probabilistic nature of cell-fate decisions is not a flaw but a feature. It enables cells with similar inputs to diverge in fate. Microheterogeneity in gene expression is now recognized as functionally significant, not mere noise,^{96,97} influencing both development and disease, including therapeutic resistance in cancer.^{98–100} Chromatin state variability—including accessibility, nucleosome positioning, and histone modifications—reflects a cell’s history and predisposes it toward particular transcriptional responses,^{96,98} especially during fate transitions involving multilineage priming. Embryonic stem cells exemplify this with substantial epigenetic variability, notably in DNA methylation, which underlies their differentiation potential.¹⁰⁰

Indeed, progenitor populations are inherently heterogeneous: individual cells show probabilistic bias toward different fates

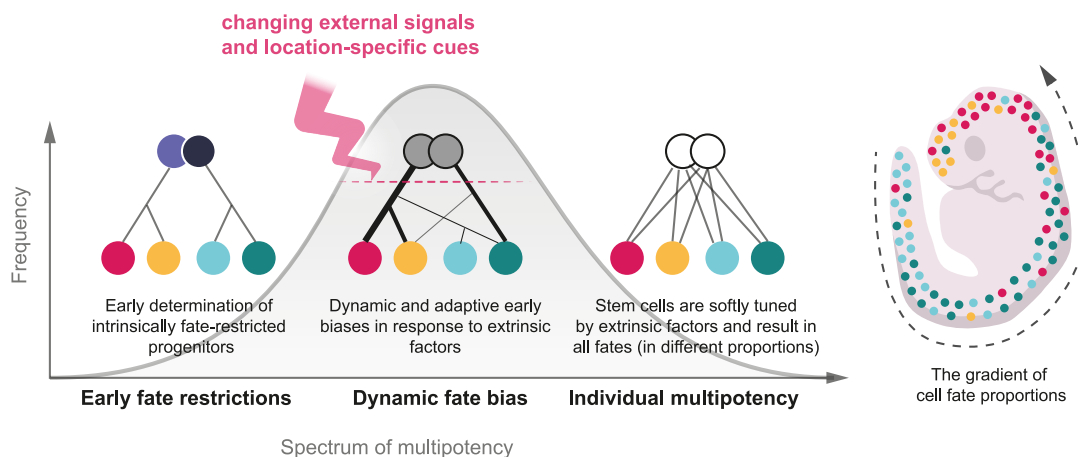
A Overview of the three main computational strategies for fate bias estimation



B Benchmarking computational strategies on scRNAseq data with clonal ground truth



C Predicted distribution of collective multipotency mechanisms



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depending on their progression within competing GRNs, even though their names (e.g., cranial neural crest) suggest a supposed monolithic nature.⁶³ This variability, though subtle on a population scale—hence “microheterogeneity”—might be crucial for balancing fate proportions. To demonstrate this logic in action, Lignell et al.¹⁰¹ combined spatially resolved single-molecule fluorescence *in situ* hybridization with machine learning to profile neural crest cells in the chick dorsal neural tube, revealing spatially organized transcriptional heterogeneity correlated with fate bias.

Other types of microheterogeneity, including asynchronous oscillations in cell cycle, transcriptional noise, senescence, and stress, also influence how cells mount a response to inductive signals.^{102–104} For instance, Kramer et al.^{105,106} explored how intrinsic states modulate signal processing in human cells. Their studies showed that cell-cycle phase affects responsiveness to short-lived signals like epidermal growth factor because mitotic chromatin compaction may hinder transcriptional responses. Since cell-cycle divergence begins early (e.g., from the 16-cell stage¹⁰⁷), subsets of cells will differ in sensitivity to morphogens at any given time. Conceptually similar mechanisms operate in other systems: in the mammalian cortex, progenitor cell-cycle phase influences cortical layer fate,¹⁰⁸ whereas in somitogenesis, oscillatory TFs determine somitic vs. intersomitic identity.^{109,110}

Importantly, the significance of microheterogeneity may lie not in its origin but in whether a cell can interpret and harness it—for instance, turning even random transcriptional noise into a cue for fate while ignoring major oscillations that carry no contextual meaning.

ADVANCES IN SINGLE-CELL TECHNOLOGIES: PROBABILISTIC MODELING OF CELL FATE AND DEVELOPMENT

The advent of single-cell technologies has provided improved resolution of the transcriptomic and epigenomic underpinnings of cell-fate decisions. These data allow us to quantitatively dissect these processes to investigate the roles of microheterogeneity, clonality, and multipotency in cell fate.^{45,102,106}

Multiple computational strategies to estimate cell-fate commitment have emerged, which attempt to model cell fate not as a hierarchical tree made of bifurcations but rather as a probabilistic process (Figure 4A). Neighborhood-based methods, exemplified by tools such as FateID, Palantir, and CellRank, leverage the cell-to-cell similarities within single-cell datasets to infer developmental trajectories and fate biases. FateID classifies fates using random forests in an iterative manner starting from terminal cells, where each iteration focuses on neighbors of the cells from the

previous iteration.¹¹¹ Fate bias is calculated as the probability from the fraction of votes for each target state, indicating likelihood of differentiation into the corresponding lineage. Palantir and CellRank both mainly rely on Markov chains constructed from directed or undirected nearest-neighbor graphs. For both tools, fate bias is calculated from random walks through the Markov chain between intermediate and terminal states.^{112,113} These methods yield a continuous measure of fate commitment, effectively mapping cells onto trajectories that can be interpreted as proxies for paths across the Waddington landscape. These methods assume a continuity of gene expression and local similarity, which is well suited to densely sampled complete datasets but can fail to recognize abrupt changes. Among the nearest-neighbor approaches, CellRank is the best-performing tool when tested on a clonally barcoded single-cell dataset accessing a fate commitment ground truth (Figure 4B). Clonal barcoding is an experimental technique for labeling cells with a unique heritable genetic barcode. This allows the tracking of cell-lineage relationships. In the present dataset, after a single barcoding event, cells were allowed to differentiate into multiple fates. This approach reveals biases in stem cell populations toward cell fates and opens the door to identifying the molecular mechanisms underlying the commitment to those fates.

Optimal transport (OT) has been proposed as an alternative to neighborhood-based computational approaches for learning fate commitment. OT is a mathematical framework that helps us understand how populations evolve over time by finding the most efficient way to transform one state distribution into another.¹¹⁴ It treats cells from a time point as a distribution and calculates the minimal “cost” of moving cells from their initial states to their final states while accounting for both distance and probability distributions. Notable existing tools are WaddingtonOT,¹¹⁵ TrajectoryNet,¹¹⁶ and MIOFlow.¹¹⁷ The strength of this strategy is that the formulation of OT applied to individual transcriptomes aligns with the principle of developmental efficiency: cells follow differentiation paths that minimize energetic or regulatory “costs.” These methods are suitable for inferring links between progenitors and progeny across multiple time points. However, the reliance on multiple time points limits the applicability of these approaches. Moreover, these tools do not provide clear advantages in terms of performance for cell-fate bias estimate recovery (Figure 4B).

Another emerging class of approaches are generative models using deep learning, such as TIGON¹¹⁸ and PRESCIENT.¹¹⁹ Both approaches rely on employing neural networks for estimating a differentiation potential gradient that can be subsequently used to generate trajectories from a given initial (real or synthetic) cell state. The biological intuition behind these two approaches

Figure 4. Fate biasing, clonality, and multipotency *in silico* and *in vivo*

- (A) Description of the three computational strategies to learn fate bias, and the possibility to use clonal barcode to improve signal.
(B) Benchmarking of multiple tools spanning the three strategies on a published single-cell RNA-seq dataset of mouse hematopoietic stem cells that were clonally barcoded and grown *in vitro* for 6 days. The resulting computationally estimated fate bias is compared to clonal ground truth. Area under the curve is a measure of performance for tool-derived fate probability vs. monocyte-neutrophil binary ground truth.
(C) The two opposing hypotheses regarding multipotency are (1) individual multipotency, whereby every stem cell can and does give rise to all possible downstream fates; and (2) early fate restriction, whereby different subsets of the stem cell pool are already primed for different lineages before lineage commitment and alternative fates are inaccessible to them. We posit a view that unifies these contrasting models, by which different stem cells do have a bias toward a certain lineage but are still capable of producing fates of other lineages given the correct stimuli.

is that these attempt to retrieve a potential cell-fate landscape, dictating to which fate a cell is likely to commit to, given its current state. Generally, the strength of deep learning lies in the flexibility of neural networks, opening new ways to learn differentiation potential and fate commitment while accounting for other co-variables such as batches and other sources of variations, as exemplified by the scVI ecosystem.¹²⁰ This flexibility, however, is not implemented in the two mentioned tools, which are also limited by the type of output these generate, which are coordinates of a synthetic trajectory in some reduced space. This means that fate commitment and relationship rely on the distances between the terminal point of a trajectory and its closest real neighbors, rendering the whole process highly dependent on parameters such as the number of simulation steps. Deep-learning-based approaches can model highly complex, dynamic processes, but assume well-sampled learning datasets covering all fates and can suffer from a lack of interpretability from the user. At the time of this review, generative modeling of trajectories does not seem to provide an obvious step up in performance for cell-fate bias estimation (Figure 4B).

Moving beyond the scope of solely transcriptomic analyses, newer experimental modalities, such as clonal lineage tracing and single-cell epigenomics (e.g., methylation profiling), are providing orthogonal layers of information that significantly enhance our understanding of fate-bias predetermination. Clonal tracing, by directly tracking the progeny of individual cells, allows for the experimental validation of predicted fate probabilities and the direct observation of lineage relationships. The experimental approach (e.g., LARRY, DARLIN) consists in introducing a barcode in each cell that will be inherited by its progeny.^{40,121} Thanks to the ability to sequence these barcodes alongside transcripts, one can computationally recover cells sharing the same barcode, indicating that they belong to the same clone. This modality enabled the emergence of computational approaches aimed at improving learning fate bias in datasets containing both transcriptome and clonal barcodes. One notable tool is Cospar, which employs a combination of OT and nearest-neighbor algorithms to build a transition map between experimental time points, taking into account the experimentally derived clonal identities.¹²² Additionally, deep-generative models were recently introduced to leverage neural networks to estimate fate bias in combination with clonal information, such as LineageVAE and DeepLineage. These approaches both rely on autoencoder (AE) architecture, which consists of learning the parameters of an encoder and decoder. The encoder takes as an input the raw transcript count matrices to learn a reduced latent space, while the decoder takes that latent space as input to reconstruct the raw data.¹²³ LineageVAE combines experimental time points and clonal identity to model a Weiner diffusion process within the latent space,¹²⁴ while DeepLineage uses long short-term memory (LSTM) within the latent space learning, considering cells and their progeny from a clone as interconnected entities.¹²⁵

Overall, while Cospar shows good performance in recovering fate bias, it is surprisingly not as capable as CellRank, which does not employ clonal information (Figure 4B). Conversely, LineageVAE does not show any clear advantage in terms of performance for fate-bias recovery. Given the high-quality clonal

data provided, these results show that, for this dataset, either the transcriptional data are sufficient to recover the correct fate bias, or the barcode-based tools exhibit limitations in how clonal information is used. A more thorough benchmark across a larger set of datasets would be necessary to clarify which scenario is more generally applicable.

In contrast, epigenomic experimental modalities provide insight into the historical and heritable component of fate bias. These multimodal approaches offer a more complete picture, integrating short-term transcriptional dynamics with longer-term lineage history and epigenetic predisposition. On one hand, chromatin accessibility enables the ability to detect the priming of regulatory elements prior to the onset of transcription. Multimodal assay for transposase-accessible chromatin with sequencing (ATAC)-RNA single-cell data show promise for improving prediction of early commitment in differentiating cells.¹²⁶ On the other hand, single-cell methylation data modalities have recently gained traction in their ability to recover clonal relationships in a non-invasive way, opening the door to the study of clonal dynamics in human tissue.¹²⁷

Despite these advancements, some general limitations remain. One is the scarcity of tools designed for advanced comparative analyses. Developmental biology is fundamentally about understanding how fate decisions are modulated by various perturbations across conditions, genotypes, or species, or in response to experimental interventions. Currently, our ability to rigorously compare fate landscapes and probabilistic commitment across such scenarios is limited. We need new computational tools that can systematically quantify and compare fate biases, identify conserved vs. divergent mechanisms of fate regulation, and evaluate the effects of perturbations on fate decisions. Developing robust methods for comparative single-cell analysis across diverse experimental contexts is crucial for translating the descriptive power of single-cell technologies into a deeper, mechanistic understanding of developmental processes.

Moreover, a common limitation across the previously described tools is the absence of an explicit low-dimensional latent space representation (such as a uniform manifold approximation and projection [UMAP]) that directly encodes cell-fate information (fate probabilities), decoupled from other sources of variation that can influence differentiation signals. This is particularly important because the microheterogeneity discussed earlier may be strongly shaped by signals such as the cell cycle, hypoxia, and metabolic rate.

Finally, a critical challenge lies in establishing causal links between specific molecular features and fate decisions. While single-cell data can reveal correlations between microheterogeneity and fate probabilities, inferring causation, particularly in the complex interplay of multiple modalities (e.g., simultaneous ATAC-seq and RNA sequencing [RNA-seq] on the same cell), remains statistically and computationally demanding. Disentangling correlation from causation requires integrated analytical frameworks that can effectively model the combined impact of gene expression, chromatin state, and lineage history on fate outcomes.

Finally, establishing causal links between specific molecular features and fate decisions will require direct perturbation. Single-cell perturbation assays such as Perturb-seq provide the

leverage to test how targeted changes reshape microheterogeneity and, in turn, shift fate probabilities across modalities.¹²⁸ Notably, CROP-seq and its recent CROPseq-multi design can natively couple perturbation with clonal tracing by repurposing internal single-guide RNA barcodes (iBARs) as high-diversity clonal barcodes, enabling lineage-aware readouts alongside CRISPR perturbations.^{129,130}

THE CONNECTION BETWEEN CELL MULTIPOTENCY AND DYNAMIC BIASING

The existence of microheterogeneity prior to fate selection challenges the reality of the classic concept of uniform multipotency. Before we go into this discussion, it is important to note that “multipotency” as a term has been somewhat vaguely used to describe cells that can give rise to cells of more than two fates.

In the history of science, multipotent cells were discovered and defined experimentally, via factual observations of cells with specific features giving rise to several different cell types *in vitro* or *in vivo*.^{131–133} This is *de facto* multipotency, whereby the cell potential becomes physically realized. However, later it became clear that multipotency might fluctuate depending on external signals and intrinsic epigenetic priming events. We use the term dynamic biasing to underscore the time-sensitive and transient nature of such regulatory states that predispose a cell toward a certain fate. Thus, multipotency started to be described in a more nuanced way, often via the potential of a cell to give rise to a set of fates and not necessarily via full realization of this potential into a physically real set of derived cell fates.¹³⁴

In addition, the term “multipotency” is often indiscriminately applied both at the population level and to individual cells. However, given the heterogeneity of multipotent cells and different degree of multipotency realization per cell, it is not obvious whether all such cells will or even can give rise to all available downstream fates. This begs the question of whether multipotency should be understood at a population or rather a single-cell level (discussed in Erickson et al.⁶⁴; Figure 4C). In the context of the neural crest, for example, the future fate-related heterogeneity within the “multipotent” population of the neural crest has been demonstrated many times. To illustrate this, seminal works from the Anderson lab suggested that Neurogenin 2 (Ngn2)-expressing neural crest cells generate both neuronal and non-neuronal lineages similar to other neural crest cells; however, the neuronal derivatives are strongly biased toward sensory neurons as opposed to autonomic neurons.¹³⁵ Interestingly, Ngn2 expression does not preclude the neural crest from becoming autonomic neurons, and thus does not constitute a hard switch of fate restriction but rather a strong probabilistic bias. Therefore, the development might globally rest on the behavior of populations of biased multipotent progenitors, each with their individual tendencies shaping clonal compositions. Such “clonal mosaics” gradually diverge, depending on place and time, which in the end leads to the vast number of fates produced at correct proportions in specific locations of the embryo.⁴⁵ In line with this, although early lineage-tracing studies showed that a single dye-labeled neural crest cell can indeed give rise to cells of most possible fates of the neural crest lineage,^{136–139}

more recent studies revealed that most of the individual neural crest cells preferentially give rise to different parts of the overarching neural crest lineage tree,^{45,134,140} thus supporting the collective multipotency model. This is not something unique, because a similar collective multipotency situation is observed in HSCs residing in a bone marrow stem cell niche.^{141,142} Furthermore, according to Konturek-Ciesla et al.,¹⁴³ the proportion of HSCs that are individually multipotent (factually give rise to a broad range of derivatives) declines with age, giving way to a collective of cells that are strongly biased. These results suggest that our hematopoiesis rather relies on the necessary collective action of HSCs when we get older.

CONCLUDING REMARKS

Multilineage priming and competing dynamics during fate selection are pervasive across various developing tissues. These phenomena likely confer flexibility in cell-fate decisions, enabling cells to adapt responsively to developmental cues and environmental factors. Together with the notion of collective multipotency, multilineage priming can explain both the robustness of development and nuances of local tissue architecture, confounded by temporal and positional signals. Further studies of collective multipotency and multilineage priming should focus on delivering an integrative approach that reconciles transcriptional and chromatin dynamics, detailed clonal information, experimentally challenged systems-level stability, and contextual modulation by signals.

ACKNOWLEDGMENTS

I.A. was supported by ERC synergy grant (KILL-OR-DIFFERENTIATE), the Swedish Research Council, the Paradifference Foundation, the Cancer Foundation in Sweden, the Knut and Alice Wallenberg Foundation, and the Austrian Science Fund (consortium SFB F78 program, Emerging Fields “Brain Resilience” consortium, and stand-alone project grants). The authors thank Dr. Olga Kharchenko for creating illustrations for this article and Dr. Johan Bostrom for proofreading.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work, the authors used ChatGPT to check the grammar and correct the excessively long and suboptimal phrasing in several places across the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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